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STATION-SIDE OPTICAL NETWORK UNIT, COMMUNICATION SYSTEM, AND CONTROL METHOD

Abstract

An aspect of the present invention is a station-side optical network unit to be connected to a subscriber-side optical network unit, the station-side optical network unit including: an acquisition unit that acquires, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; a deriving unit that derives a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired by the acquisition unit; and a setting unit that sets a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived by the deriving unit indicates a predetermined quality.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a technique of a station-side optical network unit, a communication system, and a control method.

BACKGROUND ART

[0002] A PON (passive optical network) has been widely commercialized as an optical access system. The PON has a point-to-multipoint topology, and a plurality of subscriber-side optical network units (ONUs) communicate with one optical network unit (OLT) via an optical splitter (see, for example, Non Patent Literature 1).

[0003] Meanwhile, in the next-generation optical access technology, application of multiplexing systems using a digital signal processing technology, such as OFDM (orthogonal frequency division multiplexing) or SCM (subcarrier multiplexing), to the PON has attracted attention. The multiplexing systems are used together with high-order modulation techniques and can be implemented by an inexpensive transceiver circuit having a narrow frequency band because of high bandwidth utilization efficiency. Therefore, further economic efficiency of the PON can be expected. Further, a dedicated subcarrier can be allocated to each user from a plurality of subcarriers, which makes it possible to flexibly support a new service (see, for example, Non Patent Literature 2).

CITATION LIST

Non Patent Literature

[0004] Non Patent Literature 1: Gijutsu kiso koza “GE-PON gijutsu”, Dai 1-kai (in Japanese) (Technical Basic Course “GE-PON Technology”, Part 1), NTT Technical Review, 2005. 8

[0005] Non Patent Literature 2: D. Qian, et al., “10.8-Gb/s OFDMA-PON Transmission Performance Study”, Proc. Conf. Optical Fiber Communication, NME6, March 2009.

SUMMARY OF INVENTION

Technical Problem

[0006] In the point-to-multipoint topology of the PON, optical loss between the OLT and each ONU is different, and the OLT generates a downlink signal by using one light source. Therefore, an SNR (signal-to-noise ratio) of a signal received by each ONU is different for each ONU, and thus a transmission distance and the number of branches of a PON system are determined based on an ONU having the smallest SNR. Powers of signals are wasted in ONUs other than the ONU having the smallest SNR, which causes limitation of the transmission distance and a power budget of the PON system.

[0007] In view of the above circumstances, an object of the present invention is to provide a technique capable of appropriately controlling power of a signal.

Solution to Problem

[0008] An aspect of the present invention is a station-side optical network unit to be connected to a subscriber-side optical network unit, the station-side optical network unit including: an acquisition unit that acquires, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; a deriving unit that derives a physical quantity indicating a

quality of the received signal from the signal-to-noise ratio acquired by the acquisition unit; and a setting unit that sets a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived by the deriving unit indicates a predetermined quality.

[0009] An aspect of the present invention is a communication system including a subscriber-side optical network unit and a station-side optical network unit to be connected to the subscriber-side optical network unit, in which: the station-side optical network unit includes an acquisition unit that acquires, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit, a deriving unit that derives a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired by the acquisition unit, and a setting unit that sets a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived by the deriving unit indicates a predetermined quality; and the subscriber-side optical network unit includes a measurement unit that measures the signal-to-noise ratio of the received signal, and a transmission unit that transmits the signal-to-noise ratio measured by the measurement unit to the station-side optical network unit.

[0010] An aspect of the present invention is a control method of a station-side optical network unit to be connected to a subscriber-side optical network unit, the control method including: an acquisition step of acquiring, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; a deriving step of deriving a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired in the acquisition step; and a setting step of setting a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived in the deriving step indicates a predetermined quality.

[0011] An aspect of the present invention is a control method of a communication system including a subscriber-side optical network unit and a station-side optical network unit to be connected to the subscriber-side optical network unit, in which: the station-side optical network unit acquires, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit, derives a physical quantity indicating a quality of the received signal from the acquired signal-to-noise ratio, and sets a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the derived physical quantity indicates a predetermined quality; and the subscriber-side optical network unit measures the signal-to-noise ratio of the received signal, and transmits the measured signal-to-noise ratio to the station-side optical network unit.

Advantageous Effects of Invention

[0012] The present invention can appropriately control power of a signal.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 shows a schematic configuration of a communication system according to the present embodiment.

[0014] FIG. 2 is a block diagram showing a configuration of an OLT in a first configuration example.

[0015] FIG. 3 is a block diagram showing a configuration of an ONU in the first configuration example.

[0016] FIG. 4 is a flowchart showing a flow of processing of an OLT.

[0017] FIG. 5 is a block diagram showing a configuration of an OLT in a second configuration example.

[0018] FIG. 6 is a block diagram showing a configuration of an ONU in the second configuration example.

[0019] FIG. 7 is a block diagram showing a configuration of an OLT in a third configuration example.

[0020] FIG. 8 is a block diagram showing a configuration of an ONU in the third configuration example.

[0021] FIG. 9 is a block diagram showing a configuration of an OLT in a fourth configuration example.

[0022] FIG. 10 is a block diagram showing a configuration of an ONU in the fourth configuration example.

DESCRIPTION OF EMBODIMENTS

[0023] Embodiments of the present invention will be described in detail with reference to the drawings.

[0024] FIG. 1 shows a schematic configuration of a communication system 1 according to the present embodiment. The communication system 1 includes a station-side optical network unit (hereinafter, referred to as “OLT”) 100, subscriber-side optical network units (hereinafter, referred to as “ONUs”) 200-1, 200-2, . . . , and 200-*n*, and an optical splitter 300. In the following description, the ONUs 200-1, 200-2, . . . , and 200-*n* will be referred to as the ONUs 200 unless otherwise distinguished.

[0025] As shown in FIG. 1, the communication system 1 according to the present embodiment is a PON (passive optical network) system, and the ONUs 200 communicate with one OLT 100 via the optical splitter 300. The communication system 1 allocates a dedicated subcarrier (*f*₁, . . . , and *f_n*) to each ONU 200.

[0026] The OLT 100 in the present embodiment sets a power of a signal transmitted by the OLT 100 to a minimum power among powers indicating a predetermined quality. For example, as shown in FIG. 1, optical powers of signals received by the ONU 200-1 and the ONU 200-*n* are substantially the same even if optical losses caused by optical fibers or the like are different, and the optical powers are powers indicating the predetermined quality. Meanwhile, regarding signal powers from the OLT 100 to the optical splitter 300, the signal power at the frequency *f*₁ is greatly different from the signal power at the frequency *f_n*. As described above, different powers are set to each ONU 200 in order to set the power of the signal transmitted by the OLT 100 to the power indicating the predetermined quality.

[0027] In the following description, a frequency allocated to an ONU 200-*k* (*k*=1 to *n*) is denoted by *f_k*.

First Configuration Example

[0028] FIG. 2 is a block diagram showing a configuration of the OLT 100 in a first configuration example of the embodiment. The first configuration example shows the configuration of the OLT 100 in a case where the present embodiment is applied to an OFDM multiplexing system. FIG. 3 is a block diagram showing a configuration of the ONU 200 in the first configuration example.

[0029] As shown in FIG. 2, the OLT 100 includes a PON frame processing unit 110, a transmission unit 120, a DAC (DA converter) 130, an optical front end unit 140, a reception unit 150, and a power coefficient calculation unit 160.

[0030] The PON frame processing unit 110 performs processing related to a frame including a signal to be transmitted to the ONU 200 and outputs the processed frame to the transmission unit 120. The PON frame processing unit 110 includes an acquisition unit 111, a deriving unit 112, and a setting unit 113. The acquisition unit 111 acquires, from the ONU 200, a signal-to-noise ratio (hereinafter, also referred to as “SNR”) of a received signal transmitted from the OLT 100 and received by the ONU 200. The deriving unit 112 derives a physical quantity indicating the quality

of the received signal from the acquired SNR. In the present embodiment, EVM (error vector magnitude) or BER (bit error rate) is used as the physical quantity, but the physical quantity is not limited thereto. The setting unit **113** sets a power of a signal transmitted by the OLT **100** to the minimum power among powers at which the physical quantity derived by the deriving unit **112** indicates the predetermined quality. As information for outputting a signal having the set power, a power coefficient corresponding to the power is held by the power coefficient calculation unit **160**. [0031] The transmission unit **120** includes a QAM (quadrature amplitude modulation) signal generation unit **121**, a power variable array **122**, IFFT (inverse fast Fourier transform) **123**, and a PS conversion unit **124**. In a case where the power of the signal transmitted by the OLT **100** is set to the minimum power among the powers at which the derived physical quantity indicates the predetermined quality, the QAM signal generation unit **121** generates a test pattern.

[0032] The QAM signal generation unit **121** generates a QAM signal corresponding to the frame output from the PON frame processing unit **110** for each ONU **200** and outputs the QAM signals to the power variable array **122**.

[0033] The power variable array **122** acquires power coefficients set in the power coefficient calculation unit **160**, converts the QAM signals into signals having powers corresponding to the acquired power coefficients, and outputs the signals to the IFFT **123**. The power coefficient calculation unit **160** calculates and holds power coefficients $P1(f1)$ to $P1(fn)$ corresponding to the respective frequencies $f1$ to fn . The power coefficients $P1(f1)$ to $P1(fn)$ having initial values are set in the power coefficient calculation unit **160** before being set by the setting unit **113**. The power coefficients having the initial values are coefficients for outputting a signal at a maximum power, for example.

[0034] The IFFT **123** performs inverse Fourier fast transform on each signal output from the power variable array **122** and outputs the signal to the PS conversion unit **124**. The PS conversion unit **124** converts a parallel signal output from the IFFT **123** into a serial signal and outputs the serial signal to the DAC **130**. The DAC **130** converts the digital serial signal into an analog signal and outputs the analog signal to the optical front end unit **140**.

[0035] The optical front end unit **140** includes a light source **141** and a photodetector **142**. The light source **141** is, for example, a laser diode and outputs a laser beam in response to the analog signal output from the DAC **130**. The photodetector **142** is, for example, a photodiode, converts an optical signal transmitted from the ONU **200** into an electrical signal, and outputs the electrical signal to the reception unit **150**. The reception unit **150** outputs the signal output from the optical front end unit **140** to the PON frame processing unit **110**.

[0036] Next, the configuration of the ONU **200** in the first configuration example will be described with reference to FIG. 3. As shown in FIG. 3, the ONU **200** includes an optical front end unit **210**, an ADC (AD converter) **220**, a reception unit **230**, a PON frame processing unit **240**, and a transmission unit **250**.

[0037] The optical front end unit **210** includes a photodetector **211** and a light source **212**. The photodetector **211** is, for example, a photodiode, converts an optical signal transmitted from the OLT **100** into an electrical signal, and outputs the electrical signal to the ADC **220**. The ADC **220** converts an analog signal output from the photodetector **211** into a digital signal and outputs the digital signal to the reception unit **230**.

[0038] The reception unit **230** includes a frame synchronization unit **231**, an SP conversion unit **232**, FFT (fast Fourier transform) **233**, a QAM signal determination unit **234**, and an SNR measurement unit **235**. The frame synchronization unit **231** detects the head of the frame from the signal output from the ADC **220**, extracts the frame, and outputs the frame to the SP conversion unit **232**. The SP conversion unit **232** converts a multiplexed serial signal output from the frame synchronization unit **231** into a parallel signal and outputs the parallel signal to the FFT **233**. The FFT **233** performs fast Fourier transform on the input signal for each of the frequencies $f1$ to fn and outputs the signal to the QAM signal determination unit **234**. The QAM signal determination unit

234 determines a corresponding symbol from the signal subjected to the fast Fourier transform and outputs the determination result to the PON frame processing unit **240**.

[0039] The SNR measurement unit **235** measures the SNR on the basis of the output from the FFT **233** and outputs the measurement result to the PON frame processing unit **240**. The PON frame processing unit **240** outputs the measured SNR to the transmission unit **250** as an uplink signal. The transmission unit **250** outputs the uplink signal to the light source **212**. The light source **212** is, for example, a laser diode and outputs a laser beam in response to the signal output from the transmission unit **250**.

[0040] FIG. **4** is a flowchart showing a flow of processing of the OLT **100** and the ONU **200**. The processing of the OLT **100** is indicated by step **S1XX**, and the processing of the ONU **200** is indicated by step **S2XX**. In FIG. **4**, the setting unit **113** of the OLT **100** sets a power coefficient having an initial value in the power coefficient calculation unit **160** (step **S101**). Next, the setting unit **113** initializes a loop counter k to 1 (step **S102**).

[0041] The OLT **100** transmits a test pattern signal to the ONU **200** corresponding to the frequency f_k (step **S103**). The SNR measurement unit **235** of the ONU **200** measures the SNR of the received signal (step **S201**). The SNR measurement unit **235** outputs the measured SNR to the PON frame processing unit **240**, and the PON frame processing unit **240** outputs the measured SNR to the transmission unit **250** as an uplink signal. The transmission unit **250** transmits the SNR to the OLT **100** (step **S202**). The acquisition unit **111** acquires, from the ONU **200**, the SNR of the received signal transmitted from the OLT **100** and received by the ONU **200** (step **S104**).

[0042] The deriving unit **112** derives a physical quantity indicating the quality of the received signal from the acquired SNR (step **S105**). The setting unit **113** determines whether or not the derived physical quantity is the minimum power among the powers indicating the predetermined quality (step **S106**). Here, when the physical quantity is denoted by BER, the predetermined quality is a quality at which the BER is 0.001 or less. The power coefficient set as the initial value is set to a value at which the BER is smaller than 0.001. Therefore, the power coefficient set as the initial value is likely to be more than necessary. Accordingly, in order to set the physical quantity derived by the deriving unit **112** to the minimum power among the powers indicating the predetermined quality, the setting unit **113** sets the power coefficient so as to reduce the power.

[0043] In step **S106**, whether or not the physical quantity is the minimum power is determined to be positive when, for example, $|\text{BER}-0.001| < \alpha$ (α is a positive number and is a value determined to be normally receivable by the ONU **200** through an experiment, for example) is satisfied.

[0044] When it is determined that the physical quantity is not the minimum power (step **S106**: NO), the setting unit **113** sets a power obtained by reducing the currently set power coefficient (step **S109**). Specifically, in order to set a power obtained by reducing the currently set power coefficient, the setting unit **113** causes the power coefficient calculation unit **160** to calculate and hold a power coefficient corresponding to the reduced power. The power coefficient calculation unit **160** reduces the power coefficient stepwise, for example, by each constant. Therefore, the power is reduced stepwise. In step **S103** described above, the test pattern signal having the reduced power is transmitted.

[0045] When it is determined that the physical quantity is the minimum power (step **S106**: YES), the setting unit **113** increments the loop counter k (step **S107**). The setting unit **113** determines whether or not the loop counter k is smaller than the number n of ONUs **200** (step **S108**). When the loop counter k is smaller than the number n of ONUs **200** (step **S108**: YES), the setting unit **113** returns to step **S103** to set a power of the ONU **200- k** corresponding to f_k . When the loop counter k is not smaller than the number n of ONUs **200** (step **S108**: NO), the setting unit **113** ends the processing because the power has been set for all the ONUs **200**.

[0046] The flowchart in FIG. **4** may be performed every predetermined period of time or in a case where the topology is changed.

[0047] As described above, the setting unit **113** sets the power of the signal transmitted by the OLT

100 to the minimum power among the powers at which the physical quantity derived by the deriving unit **112** indicates the predetermined quality. This makes it possible to suppress waste of power while maintaining the predetermined quality for each ONU **200**. Therefore, the power of the signal can be appropriately controlled.

Second Configuration Example

[0048] FIG. 5 is a block diagram showing a configuration of an OLT **100** in a second configuration example of the embodiment. The second configuration example shows the configuration of the OLT **100** in a case where the present embodiment is applied to an SCM multiplexing system. FIG. 6 is a block diagram showing a configuration of an ONU **1200** in the second configuration example.

[0049] As shown in FIG. 5, the OLT **100** includes a PON frame processing unit **110**, a transmission unit **120**, a DAC (DA converter) **130**, an optical front end unit **140**, a reception unit **150**, and a power coefficient calculation unit **160**.

[0050] The PON frame processing unit **110** performs processing related to a frame including a signal to be transmitted to the ONU **1200** and outputs the processed frame to the transmission unit **120**. The PON frame processing unit **110** includes an acquisition unit **111**, a deriving unit **112**, and a setting unit **113**. The acquisition unit **111** acquires, from the ONU **1200**, the SNR of a received signal transmitted from the OLT **100** and received by the ONU **1200**. The deriving unit **112** derives a physical quantity indicating the quality of the received signal from the acquired SNR. In the present embodiment, the EVM or BER is used as the physical quantity, but the physical quantity is not limited thereto. The setting unit **113** sets a power of a signal transmitted by the OLT **100** to the minimum power among powers at which the physical quantity derived by the deriving unit **112** indicates the predetermined quality. As information for outputting a signal having the set power, a power coefficient corresponding to the power is held by the power coefficient calculation unit **160**.

[0051] The transmission unit **120** includes a QAM signal generation unit **121**, IQ mixers **125**, a power variable array **122**, and a power combiner **126**. In a case where the power of the signal transmitted by the OLT **100** is set to the minimum power among the powers at which the derived physical quantity indicates the predetermined quality, the QAM signal generation unit **121** generates a test pattern.

[0052] The QAM signal generation unit **121** generates a QAM signal corresponding to the frame output from the PON frame processing unit **110** for each ONU **1200** and outputs the QAM signal to each IQ mixer **125**. The IQ mixer **125** is provided for each of the frequencies f_1 to f_n . The IQ mixer **125** modulates the QAM signal and outputs the modulated QAM signal to the power variable array **122**.

[0053] The power variable array **122** acquires power coefficients set in the power coefficient calculation unit **160**, converts the signals output from the IQ mixers **125** into signals having powers corresponding to the acquired power coefficients, and outputs the signals to the power combiner **126**. The power coefficient calculation unit **160** calculates and holds the power coefficients $P_1(f_1)$ to $P_1(f_n)$ corresponding to the respective frequencies f_1 to f_n . The power coefficients $P_1(f_1)$ to $P_1(f_n)$ having initial values are set in the power coefficient calculation unit **160** before being set by the setting unit **113**. The power coefficients having the initial values are coefficients for outputting a signal at a maximum power, for example.

[0054] The power combiner **126** combines the signals output from the power variable array **122** and outputs the combined signal to the DAC **130**. The DAC **130** converts the signal output from the power combiner **126** into an analog signal and outputs the analog signal to the optical front end unit **140**.

[0055] The optical front end unit **140** includes a light source **141** and a photodetector **142**. The light source **141** is, for example, a laser diode and outputs a laser beam in response to the analog signal output from the DAC **130**. The photodetector **142** is, for example, a photodiode, converts

an optical signal transmitted from the ONU **1200** into an electrical signal, and outputs the electrical signal to the reception unit **1150**. The reception unit **1150** outputs the signal output from the optical front end unit **1140** to the PON frame processing unit **1110**.

[0056] Next, the configuration of the ONU **1200** in the second configuration example will be described with reference to FIG. **6**. As shown in FIG. **6**, the ONU **1200** includes an optical front end unit **1210**, an ADC (AD converter) **1220**, a reception unit **1230**, a PON frame processing unit **1240**, and a transmission unit **1250**.

[0057] The optical front end unit **1210** includes a photodetector **1211** and a light source **1212**. The photodetector **1211** is, for example, a photodiode, converts an optical signal transmitted from the OLT **1100** into an electrical signal, and outputs the electrical signal to the ADC **1220**. The ADC **1220** converts an analog signal output from the photodetector **1211** into a digital signal and outputs the digital signal to the reception unit **1230**.

[0058] The reception unit **1230** includes a frame synchronization unit **1231**, a power combiner **1236**, IQ mixers **1237**, a QAM signal determination unit **1234**, and an SNR measurement unit **1235**. The frame synchronization unit **1231** detects the head of the frame from the signal output from the ADC **1220**, extracts the frame, and outputs the frame to the power combiner **1236**. The power combiner **1236** demultiplexes the signal output from the frame synchronization unit **1231** and outputs the demultiplexed signals to the IQ mixers **1237**. The IQ mixer **1237** is provided for each of the frequencies f_1 to f_N . The IQ mixer **1237** demodulates the input signal and outputs the demodulated signal to the QAM signal determination unit **1234**. The QAM signal determination unit **1234** determines a corresponding symbol on the basis of the signal demodulated by the IQ mixer **1237** and outputs the determination result to the PON frame processing unit **1240**.

[0059] The SNR measurement unit **1235** measures the SNR on the basis of the output from the IQ mixer **1237** and outputs the measurement result to the PON frame processing unit **1240**. The PON frame processing unit **1240** outputs the measured SNR to the transmission unit **1250** as an uplink signal. The transmission unit **1250** outputs the uplink signal to the light source **1212**. The light source **1212** is, for example, a laser diode and outputs a laser beam in response to the signal output from the transmission unit **1250**.

[0060] Also in the second configuration example, it is possible to suppress waste of power while maintaining the predetermined quality for each ONU by processing similar to the processing in FIG. **4**. Therefore, the power of the signal can be appropriately controlled.

[0061] A flow of the processing in the second configuration example will be described by diverting the flowchart of FIG. **4**. In FIG. **4**, the setting unit **1113** of the OLT **1100** sets a power coefficient having an initial value in the power coefficient calculation unit **1160** (step **S101**). Next, the setting unit **1113** initializes the loop counter k to 1 (step **S102**).

[0062] The OLT **1100** transmits a test pattern signal to the ONU **1200** corresponding to the frequency f_k (step **S103**). The SNR measurement unit **1235** of the ONU **1200** measures the SNR of the received signal (step **S201**). The SNR measurement unit **1235** outputs the measured SNR to the PON frame processing unit **1240**, and the PON frame processing unit **1240** outputs the measured SNR to the transmission unit **1250** as an uplink signal. The transmission unit **1250** transmits the SNR to the OLT **1100** (step **S202**). The acquisition unit **1111** acquires, from the ONU **1200**, the SNR of the received signal transmitted from the OLT **1100** and received by the ONU **1200** (step **S104**).

[0063] The deriving unit **1112** derives a physical quantity indicating the quality of the received signal from the acquired SNR (step **S105**). The setting unit **1113** determines whether or not the derived physical quantity is the minimum power among the powers indicating the predetermined quality (step **S106**). Here, when the physical quantity is denoted by BER, the predetermined quality is a quality at which the BER is 0.001 or less. The power coefficient set as the initial value is set to a value at which the BER is smaller than 0.001. Therefore, the power coefficient set as the initial value is likely to be more than necessary. Accordingly, in order to set the physical quantity derived

by the deriving unit **1112** to the minimum power among the powers indicating the predetermined quality, the setting unit **1113** sets the power coefficient so as to reduce the power.

[0064] In step **S106**, whether or not the physical quantity is the minimum power is determined to be positive when, for example, $|\text{BER}-0.001| < \alpha$ (α is a positive number and is a value determined to be normally receivable by the ONU **1200** through an experiment, for example) is satisfied.

[0065] When it is determined that the physical quantity is not the minimum power (step **S106**: NO), the setting unit **1113** sets a power obtained by reducing the currently set power coefficient (step **S109**). Specifically, in order to set a power obtained by reducing the currently set power coefficient, the setting unit **1113** causes the power coefficient calculation unit **1160** to calculate and hold a power coefficient corresponding to the reduced power. The power coefficient calculation unit **1160** reduces the power coefficient stepwise, for example, by each constant. Therefore, the power is reduced stepwise. In step **S103** described above, the test pattern signal having the reduced power is transmitted.

[0066] When it is determined that the physical quantity is the minimum power (step **S106**: YES), the setting unit **1113** increments the loop counter k (step **S107**). The setting unit **1113** determines whether or not the loop counter k is equal to or smaller than the number n of ONUs **1200** (step **S108**). When the loop counter k is equal to or smaller than the number n of ONUs **1200** (step **S108**: YES), the setting unit **1113** returns to step **S103** to set a power of an ONU **1200**- k corresponding to f_k . When the loop counter k is larger than the number n of ONUs **1200** (step **S108**: NO), the setting unit **1113** ends the processing because the power has been set for all the ONUs **1200**.

[0067] In the second configuration example, the flowchart in FIG. 4 may be performed every predetermined period of time or in a case where the topology is changed.

[0068] As described above, the setting unit **1113** sets the power of the signal transmitted by the OLT **1100** to the minimum power among the powers at which the physical quantity derived by the deriving unit **1112** indicates the predetermined quality. This makes it possible to suppress waste of power while maintaining the predetermined quality for each ONU **1200**. Therefore, the power of the signal can be appropriately controlled.

Third Configuration Example

[0069] FIG. 7 is a block diagram showing a configuration of an OLT **2100** in a third configuration example of the embodiment. The third configuration example shows the configuration of the OLT **2100** in a case where the present embodiment is applied to an SSB (single sideband)-SCM multiplexing system. FIG. 8 is a block diagram showing a configuration of an ONU **2200** in the third configuration example.

[0070] As shown in FIG. 7, the OLT **2100** includes a PON frame processing unit **2110**, a transmission unit **2120**, DACs (DA converters) **2131** and **2132**, an optical front end unit **2140**, a reception unit **2150**, and a power coefficient calculation unit **2160**. The PON frame processing unit **2110** performs processing related to a frame including a signal to be transmitted to the ONU **2200** and outputs the processed frame to the transmission unit **2120**. The PON frame processing unit **2110** includes an acquisition unit **2111**, a deriving unit **2112**, and a setting unit **2113**. The acquisition unit **2111** acquires, from the ONU **2200**, the SNR of a received signal transmitted from the OLT **2100** and received by the ONU **2200**. The deriving unit **2112** derives a physical quantity indicating the quality of the received signal from the acquired SNR. In the present embodiment, the EVM or BER is used as the physical quantity, but the physical quantity is not limited thereto. The setting unit **2113** sets a power of a signal transmitted by the OLT **2100** to the minimum power among powers at which the physical quantity derived by the deriving unit **2112** indicates the predetermined quality. As information for outputting a signal having the set power, a power coefficient corresponding to the power is held by the power coefficient calculation unit **2160**.

[0071] The transmission unit **2120** includes a QAM signal generation unit **2121**, IQ mixers **2125**, a power variable array **2122**, and a power combiner **2126**. In a case where the power of the signal transmitted by the OLT **2100** is set to the minimum power among the powers at which the derived

physical quantity indicates the predetermined quality, the QAM signal generation unit **2121** generates a test pattern.

[0072] The QAM signal generation unit **2121** generates a QAM signal corresponding to the frame output from the PON frame processing unit **2110** for each ONU **2200** and outputs the QAM signal to each IQ mixer **2125**. The IQ mixer **2125** is provided for each of the frequencies f_1 to f_N . The IQ mixer **2125** modulates the QAM signal and outputs the modulated QAM signal to the power variable array **2122**.

[0073] The power variable array **2122** acquires power coefficients set in the power coefficient calculation unit **2160**, converts the signals output from the IQ mixers **2125** into signals having powers corresponding to the acquired power coefficients, and outputs the signals to the power combiner **2126**. The power coefficient calculation unit **2160** calculates and holds the power coefficients $P_1(f_1)$ to $P_1(f_n)$ corresponding to the respective frequencies f_1 to f_n . The power coefficients $P_1(f_1)$ to $P_1(f_n)$ having initial values are set in the power coefficient calculation unit **2160** before being set by the setting unit **2113**. The power coefficients having the initial values are coefficients for outputting a signal at a maximum power, for example.

[0074] The power combiner **2126** combines the signals output from the power variable array **2122**, outputs an in-phase component signal to the DAC **2131**, and outputs a quadrature component signal to the DAC **2132**. The DACs **2131** and **2132** convert the signals output from the power combiner **2126** into analog signals and output the analog signals to the optical front end unit **2140**.

[0075] The optical front end unit **2140** includes a light source **2141**, an IQ modulator **2143**, and a photodetector **2142**. The light source **2141** is, for example, a laser diode. The IQ modulator **2143** receives the signals input from the DACs **2131** and **2132** and modulates and outputs laser beams output from the light source **2141** in response to the input signals. The photodetector **2142** is, for example, a photodiode, converts an optical signal transmitted from the ONU **2200** into an electrical signal, and outputs the electrical signal to the reception unit **2150**. The reception unit **2150** outputs the signal output from the optical front end unit **2140** to the PON frame processing unit **2110**.

[0076] Next, the configuration of the ONU **2200** in the third configuration example will be described with reference to FIG. 8. As shown in FIG. 8, the ONU **2200** includes an optical front end unit **2210**, an ADC (AD converter) **2220**, a reception unit **2230**, a PON frame processing unit **2240**, and a transmission unit **2250**.

[0077] The optical front end unit **2210** includes a photodetector **2211** and a light source **2212**. The photodetector **2211** is, for example, a photodiode, converts an optical signal transmitted from the OLT **2100** into an electrical signal, and outputs the electrical signal to the ADC **2220**. The ADC **2220** converts an analog signal output from the photodetector **2211** into a digital signal and outputs the digital signal to the reception unit **2230**.

[0078] The reception unit **2230** includes a frame synchronization unit **2231**, a power combiner **2236**, IQ mixers **2237**, a QAM signal determination unit **2234**, and an SNR measurement unit **2235**. The frame synchronization unit **2231** detects the head of the frame from the signal output from the ADC **2220**, extracts the frame, and outputs the frame to the power combiner **2236**. The power combiner **2236** demultiplexes the signal output from the frame synchronization unit **2231** and outputs the demultiplexed signals to the IQ mixers **2237**. The IQ mixer **2237** is provided for each of the frequencies f_1 to f_N . The IQ mixer **2237** demodulates the input signal and outputs the demodulated signal to the QAM signal determination unit **2234**. The QAM signal determination unit **2234** determines a corresponding symbol on the basis of the signal demodulated by the IQ mixer **2237** and outputs the determination result to the PON frame processing unit **2240**.

[0079] The SNR measurement unit **2235** measures the SNR on the basis of the output from the IQ mixer **2237** and outputs the measurement result to the PON frame processing unit **2240**. The PON frame processing unit **2240** outputs the measured SNR to the transmission unit **2250** as an uplink signal. The transmission unit **2250** outputs the uplink signal to the light source **2212**. The light source **2212** is, for example, a laser diode and outputs a laser beam in response to the signal output

from the transmission unit **2250**.

[0080] Also in the third configuration example, it is possible to suppress waste of power while maintaining the predetermined quality for each ONU by processing similar to the processing in FIG. **4**. Therefore, the power of the signal can be appropriately controlled.

[0081] A flow of the processing in the third configuration example will be described by diverting the flowchart of FIG. **4**. In FIG. **4**, the setting unit **2113** of the OLT **2100** sets a power coefficient having an initial value in the power coefficient calculation unit **2160** (step **S101**). Next, the setting unit **2113** initializes the loop counter k to 1 (step **S102**).

[0082] The OLT **2100** transmits a test pattern signal to the ONU **2200** corresponding to the frequency f_k (step **S103**). The SNR measurement unit **2235** of the ONU **2200** measures the SNR of the received signal (step **S201**). The SNR measurement unit **2235** outputs the measured SNR to the PON frame processing unit **2240**, and the PON frame processing unit **2240** outputs the measured SNR to the transmission unit **2250** as an uplink signal. The transmission unit **2250** transmits the SNR to the OLT **2100** (step **S202**). The acquisition unit **2111** acquires, from the ONU **2200**, the SNR of the received signal transmitted from the OLT **2100** and received by the ONU **2200** (step **S104**).

[0083] The deriving unit **2112** derives a physical quantity indicating the quality of the received signal from the acquired SNR (step **S105**). The setting unit **2113** determines whether or not the derived physical quantity is the minimum power among the powers indicating the predetermined quality (step **S106**). Here, when the physical quantity is denoted by BER, the predetermined quality is a quality at which the BER is 0.001 or less. The power coefficient set as the initial value is set to a value at which the BER is smaller than 0.001. Therefore, the power coefficient set as the initial value is likely to be more than necessary. Accordingly, in order to set the physical quantity derived by the deriving unit **2112** to the minimum power among the powers indicating the predetermined quality, the setting unit **2113** sets the power coefficient so as to reduce the power.

[0084] In step **S106**, whether or not the physical quantity is the minimum power is determined to be positive when, for example, $|\text{BER}-0.001|<\alpha$ (α is a positive number and is a value determined to be normally receivable by the ONU **2200** through an experiment, for example) is satisfied.

[0085] When it is determined that the physical quantity is not the minimum power (step **S106**: NO), the setting unit **2113** sets a power obtained by reducing the currently set power coefficient (step **S109**). Specifically, in order to set a power obtained by reducing the currently set power coefficient, the setting unit **2113** causes the power coefficient calculation unit **2160** to calculate and hold a power coefficient corresponding to the reduced power. The power coefficient calculation unit **2160** reduces the power coefficient stepwise, for example, by each constant. Therefore, the power is reduced stepwise. In step **S103** described above, the test pattern signal having the reduced power is transmitted.

[0086] When it is determined that the physical quantity is the minimum power (step **S106**: YES), the setting unit **2113** increments the loop counter k (step **S107**). The setting unit **2113** determines whether or not the loop counter k is smaller than the number n of ONUs **2200** (step **S108**). When the loop counter k is smaller than the number n of ONUs **2200** (step **S108**: YES), the setting unit **2113** returns to step **S103** to set a power of an ONU **2200**- k corresponding to f_k . When the loop counter k is not smaller than the number n of ONUs **2200** (step **S108**: NO), the setting unit **2113** ends the processing because the power has been set for all the ONUs **2200**.

[0087] In the third configuration example, the flowchart in FIG. **4** may be performed every predetermined period of time or in a case where the topology is changed.

[0088] As described above, the setting unit **2113** sets the power of the signal transmitted by the OLT **2100** to the minimum power among the powers at which the physical quantity derived by the deriving unit **2112** indicates the predetermined quality. This makes it possible to suppress waste of power while maintaining the predetermined quality for each ONU **2200**. Therefore, the power of the signal can be appropriately controlled.

Fourth Configuration Example

[0089] FIG. **9** is a block diagram showing a configuration of an OLT **3100** in a fourth configuration example of the embodiment. The fourth configuration example shows the configuration of the OLT **3100** in a case where the present embodiment is applied to the SSB-SCM multiplexing system. FIG. **10** is a block diagram showing a configuration of an ONU **3200** including a coherent optical detector in the fourth configuration example.

[0090] As shown in FIG. **9**, the OLT **3100** includes a PON frame processing unit **3110**, a transmission unit **3120**, DACs (DA converters) **3131** and **3132**, an optical front end unit **3140**, a reception unit **3150**, and a power coefficient calculation unit **3160**.

[0091] The PON frame processing unit **3110** performs processing related to a frame including a signal to be transmitted to the ONU **3200** and outputs the processed frame to the transmission unit **3120**. The PON frame processing unit **3110** includes an acquisition unit **3111**, a deriving unit **3112**, and a setting unit **3113**. The acquisition unit **3111** acquires, from the ONU **3200**, the SNR of a received signal transmitted from the OLT **3100** and received by the ONU **3200**. The deriving unit **3112** derives a physical quantity indicating the quality of the received signal from the acquired SNR. In the present embodiment, the EVM or BER is used as the physical quantity, but the physical quantity is not limited thereto. The setting unit **3113** sets a power of a signal transmitted by the OLT **3100** to the minimum power among powers at which the physical quantity derived by the deriving unit **3112** indicates the predetermined quality. As information for outputting a signal having the set power, a power coefficient corresponding to the power is held by the power coefficient calculation unit **3160**.

[0092] The transmission unit **3120** includes a QAM signal generation unit **3121**, IQ mixers **3125**, a power variable array **3122**, and a power combiner **3126**. In a case where the power of the signal transmitted by the OLT **3100** is set to the minimum power among the powers at which the derived physical quantity indicates the predetermined quality, the QAM signal generation unit **3121** generates a test pattern.

[0093] The QAM signal generation unit **3121** generates a QAM signal corresponding to the frame output from the PON frame processing unit **3110** for each ONU **3200** and outputs the QAM signal to each IQ mixer **3125**. The IQ mixer **3125** is provided for each of the frequencies f_1 to f_N . The IQ mixer **3125** modulates the QAM signal and outputs the modulated QAM signal to the power variable array **3122**.

[0094] The power variable array **3122** acquires power coefficients set in the power coefficient calculation unit **3160**, converts the signals output from the IQ mixers **3125** into signals having powers corresponding to the acquired power coefficients, and outputs the signals to the power combiner **3126**. The power coefficient calculation unit **3160** calculates and holds the power coefficients $P_1(f_1)$ to $P_1(f_N)$ corresponding to the respective frequencies f_1 to f_N . The power coefficients $P_1(f_1)$ to $P_1(f_N)$ having initial values are set in the power coefficient calculation unit **3160** before being set by the setting unit **3113**. The power coefficients having the initial values are coefficients for outputting a signal at a maximum power, for example.

[0095] The power combiner **3126** combines the signals output from the power variable array **3122**, outputs an in-phase component signal to the DAC **3131**, and outputs a quadrature component signal to the DAC **3132**. The DACs **3131** and **3132** convert the signals output from the power combiner **3126** into analog signals and output the analog signals to the optical front end unit **3140**.

[0096] The optical front end unit **3140** includes a light source **3141**, an IQ modulator **3143**, and a photodetector **3142**. The light source **3141** is, for example, a laser diode. The IQ modulator **3143** receives the signals input from the DACs **3131** and **3132** and modulates and outputs laser beams output from the light source **3141** in response to the input signals. The photodetector **3142** is, for example, a photodiode, converts an optical signal transmitted from the ONU **3200** into an electrical signal, and outputs the electrical signal to the reception unit **3150**. The reception unit **3150** outputs the signal output from the optical front end unit **3140** to the PON frame processing unit **3110**.

[0097] Next, the configuration of the ONU **3200** in the fourth configuration example will be described with reference to FIG. **10**. As shown in FIG. **10**, the ONU **3200** includes an optical front end unit **3210**, ADCs (AD converters) **3221** and **3222**, a reception unit **3230**, a PON frame processing unit **3240**, and a transmission unit **3250**.

[0098] The optical front end unit **3210** includes a coherent optical detector **3213**, a local oscillation light source **3214**, and a light source **3212**. The coherent optical detector **3213** detects an in-phase component signal and a quadrature component signal by causing a laser beam from the local oscillation light source **3214** to interfere with the received optical signal. The detected in-phase component signal is output to the ADC **3221**. The detected quadrature component signal is output to the ADC **3222**.

[0099] The ADC **3221** converts the in-phase component signal output from the coherent optical detector **3213** into a digital signal and outputs the digital signal to the reception unit **3230**. The ADC **3222** converts the quadrature component signal output from the coherent optical detector **3213** into a digital signal and outputs the digital signal to the reception unit **3230**.

[0100] The reception unit **3230** includes a frame synchronization unit **3231**, a power combiner **3236**, IQ mixers **3237**, a QAM signal determination unit **3234**, and an SNR measurement unit **3235**. The frame synchronization unit **3231** detects the head of the frames from the signals output from the ADCs **3221** and **3222**, extracts the frames, and outputs the frames to the power combiner **3236**. The power combiner **3236** demultiplexes the signal output from the frame synchronization unit **3231** and outputs the demultiplexed signals to the IQ mixers **3237**. The IQ mixer **3237** is provided for each of the frequencies f_1 to f_N . The IQ mixer **3237** demodulates the input signal and outputs the demodulated signal to the QAM signal determination unit **3234**. The QAM signal determination unit **3234** determines a corresponding symbol on the basis of the signal demodulated by the IQ mixer **3237** and outputs the determination result to the PON frame processing unit **3240**.

[0101] The SNR measurement unit **3235** measures the SNR on the basis of the output from the IQ mixer **3237** and outputs the measurement result to the PON frame processing unit **3240**. The PON frame processing unit **3240** outputs the measured SNR to the transmission unit **3250** as an uplink signal. The transmission unit **3250** outputs the uplink signal to the light source **3212**. The light source **3212** is, for example, a laser diode and outputs a laser beam in response to the signal output from the transmission unit **3250**.

[0102] Also in the fourth configuration example, it is possible to suppress waste of power while maintaining the predetermined quality for each ONU by processing similar to the processing in FIG. **4**. Therefore, the power of the signal can be appropriately controlled.

[0103] A flow of the processing in the fourth configuration example will be described by diverting the flowchart of FIG. **4**. In FIG. **4**, the setting unit **3113** of the OLT **3100** sets a power coefficient having an initial value in the power coefficient calculation unit **3160** (step **S101**). Next, the setting unit **3113** initializes the loop counter k to 1 (step **S102**).

[0104] The OLT **3100** transmits a test pattern signal to the ONU **3200** corresponding to the frequency f_k (step **S103**). The SNR measurement unit **3235** of the ONU **3200** measures the SNR of the received signal (step **S201**). The SNR measurement unit **3235** outputs the measured SNR to the PON frame processing unit **3240**, and the PON frame processing unit **3240** outputs the measured SNR to the transmission unit **3250** as an uplink signal. The transmission unit **3250** transmits the SNR to the OLT **3100** (step **S202**). The acquisition unit **3111** acquires, from the ONU **3200**, the SNR of the received signal transmitted from the OLT **3100** and received by the ONU **3200** (step **S104**).

[0105] The deriving unit **3112** derives a physical quantity indicating the quality of the received signal from the acquired SNR (step **S105**). The setting unit **3113** determines whether or not the derived physical quantity is the minimum power among the powers indicating the predetermined quality (step **S106**). Here, when the physical quantity is denoted by BER, the predetermined quality is a quality at which the BER is 0.001 or less. The power coefficient set as the initial value is set to

a value at which the BER is smaller than 0.001. Therefore, the power coefficient set as the initial value is likely to be more than necessary. Accordingly, in order to set the physical quantity derived by the deriving unit **3112** to the minimum power among the powers indicating the predetermined quality, the setting unit **3113** sets the power coefficient so as to reduce the power.

[0106] In step **S106**, whether or not the physical quantity is the minimum power is determined to be positive when, for example, $|\text{BER}-0.001|<\alpha$ (α is a positive number and is a value determined to be normally receivable by the ONU **3200** through an experiment, for example) is satisfied.

[0107] When it is determined that the physical quantity is not the minimum power (step **S106**: NO), the setting unit **3113** sets a power obtained by reducing the currently set power coefficient (step **S109**). Specifically, in order to set a power obtained by reducing the currently set power coefficient, the setting unit **3113** causes the power coefficient calculation unit **3160** to calculate and hold a power coefficient corresponding to the reduced power. The power coefficient calculation unit **3160** reduces the power coefficient stepwise, for example, by each constant. Therefore, the power is reduced stepwise. In step **S103** described above, the test pattern signal having the reduced power is transmitted.

[0108] When it is determined that the physical quantity is the minimum power (step **S106**: YES), the setting unit **3113** increments the loop counter k (step **S107**). The setting unit **3113** determines whether or not the loop counter k is equal to or smaller than the number n of ONUs **3200** (step **S108**). When the loop counter k is equal to or smaller than the number n of ONUs **3200** (step **S108**: YES), the setting unit **3113** returns to step **S103** to set a power of an ONU **3200- k** corresponding to f_k . When the loop counter k is larger than the number n of ONUs **3200** (step **S108**: NO), the setting unit **3113** ends the processing because the power has been set for all the ONUs **3200**.

[0109] In the fourth configuration example, the flowchart in FIG. **4** may be performed every predetermined period of time or in a case where the topology is changed.

[0110] As described above, the setting unit **3113** sets the power of the signal transmitted by the OLT **3100** to the minimum power among the powers at which the physical quantity derived by the deriving unit **3112** indicates the predetermined quality. This makes it possible to suppress waste of power while maintaining the predetermined quality for each ONU **3200**. Therefore, the power of the signal can be appropriately controlled.

[0111] In the embodiment described above, the power coefficient is reduced stepwise by each constant, but each constant may be a minimum unit that can be reduced. In any case, the power of the signal transmitted by the OLT is set to the minimum power among the powers at which the physical quantity indicates the predetermined quality from among the powers reduced stepwise.

[0112] The PON frame processing units **110**, **240**, **1110**, **1240**, **2110**, **2240**, **3110**, and **3240**, the transmission units **120**, **1120**, **2120**, and **3120**, the reception units **230**, **1230**, **2230**, and **3230**, and the power coefficient calculation units **160**, **1160**, **2160**, and **3160** may be configured by using a processor such as a central processing unit (CPU) and a memory. In this case, the PON frame processing units **110**, **240**, **1110**, **1240**, **2110**, **2240**, **3110**, and **3240**, the transmission units **120**, **1120**, **2120**, and **3120**, the reception units **230**, **1230**, **2230**, and **3230**, and the power coefficient calculation units **160**, **1160**, **2160**, and **3160** function as the PON frame processing units **110**, **240**, **1110**, **1240**, **2110**, **2240**, **3110**, and **3240**, the transmission units **120**, **1120**, **2120**, and **3120**, the reception units **230**, **1230**, **2230**, and **3230**, and the power coefficient calculation units **160**, **1160**, **2160**, and **3160** by the processor executing a program. All or some of the functions of the PON frame processing units **110**, **240**, **1110**, **1240**, **2110**, **2240**, **3110**, and **3240**, the transmission units **120**, **1120**, **2120**, and **3120**, the reception units **230**, **1230**, **2230**, and **3230**, and the power coefficient calculation units **160**, **1160**, **2160**, and **3160** may be implemented by using hardware such as an application specific integrated circuit (ASIC), a programmable logic device (PLD), or a field programmable gate array (FPGA). The above program may be recorded in a computer-readable recording medium. The computer-readable recording medium is, for example, a portable medium such as a flexible disk, a magneto-optical disc, a ROM, a CD-ROM, or a semiconductor

storage device (e.g., a solid state drive (SSD)) or a storage device such as a hard disk or a semiconductor storage device built in a computer system. The program may be transmitted via a telecommunication line.

[0113] Although the embodiments of the present invention have been described in detail with reference to the drawings, specific configurations are not limited to the embodiments, and include design and the like within the scope of the present invention without departing from the gist of the present invention.

Industrial Applicability

[0114] The present invention is applicable to a communication system that performs communication through an optical fiber transmission line.

REFERENCE SIGNS LIST

[0115] **1** . . . Communication system [0116] **100, 1100, 2100, 3100** . . . OLT [0117] **200, 1200, 2200, 3200** . . . ONU [0118] **111, 1111, 2111, 3111** . . . Acquisition unit [0119] **112, 1112, 2112, 3112** . . . Deriving unit [0120] **113, 1113, 2113, 3113** . . . Setting unit

Claims

1. A station-side optical network unit to be connected to a subscriber-side optical network unit, the station-side optical network unit comprising: a processor; and a storage medium having computer program instructions stored thereon, when executed by the processor, perform to: acquire, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; derive a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired; and set a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived indicates a predetermined quality.
2. The station-side optical network unit according to claim 1, wherein in a case where a plurality of the subscriber-side optical network units is connected, the processor sets the power of the signal transmitted by the subscriber-side optical network unit for each of the plurality of subscriber-side optical network units.
3. The station-side optical network unit according to claim 1, wherein the processor sets the power of the signal transmitted by the station-side optical network unit to the minimum power among the powers at which the physical quantity derived indicates the predetermined quality by reducing stepwise a power determined as an initial value.
4. A communication system comprising a subscriber-side optical network unit and a station-side optical network unit to be connected to the subscriber-side optical network unit, wherein the station-side optical network unit comprises a first processor; and a first storage medium having computer program instructions stored thereon, when executed by the first processor, perform to: acquire, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit, derive a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired, and set a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived indicates a predetermined quality; and the subscriber-side optical network unit comprises a second processor; and a second storage medium having computer program instructions stored thereon, when executed by the second processor, perform to: measure the signal-to-noise ratio of the received signal, and transmit the signal-to-noise ratio measured to the station-side optical network unit.
5. The communication system according to claim 4, wherein in a case where a plurality of the subscriber-side optical network units is connected, the first processor sets the power of the signal transmitted by the subscriber-side optical network unit for each of the plurality of subscriber-side

optical network units.

6. The communication system according to claim 4, wherein the first processor sets the power of the signal transmitted by the station-side optical network unit to the minimum power among the powers at which the physical quantity derived indicates the predetermined quality by reducing stepwise a power determined as an initial value.

7. A control method of a station-side optical network unit to be connected to a subscriber-side optical network unit, the control method comprising: acquiring, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; deriving a physical quantity indicating a quality of the received signal from the signal-to-noise ratio acquired; and setting a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the physical quantity derived indicates a predetermined quality.

8. A control method of a communication system including a subscriber-side optical network unit and a station-side optical network unit to be connected to the subscriber-side optical network unit, the control method comprising: acquiring, by the station-side optical network unit, from the subscriber-side optical network unit, a signal-to-noise ratio of a received signal transmitted from the station-side optical network unit and received by the subscriber-side optical network unit; deriving, by the station-side optical network unit, a physical quantity indicating a quality of the received signal from the acquired signal-to-noise ratio; setting, by the station-side optical network unit, a power of a signal transmitted by the station-side optical network unit to a minimum power among powers at which the derived physical quantity indicates a predetermined quality; measuring, by the optical network unit, the signal-to-noise ratio of the received signal, and transmitting, by the optical network unit, the measured signal-to-noise ratio to the station-side optical network unit.

9. The station-side optical network unit according to claim 2, wherein the processor sets the power of the signal transmitted by the station-side optical network unit to the minimum power among the powers at which the physical quantity derived indicates the predetermined quality by reducing stepwise a power determined as an initial value.

10. The communication system according to claim 5, wherein the first processor sets the power of the signal transmitted by the station-side optical network unit to the minimum power among the powers at which the physical quantity derived indicates the predetermined quality by reducing stepwise a power determined as an initial value.
